

11

$x_1^{l-1} f$

1

1

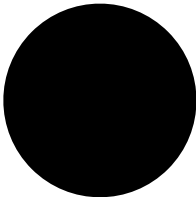
11

$x_1^l f$

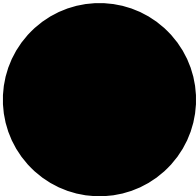
1

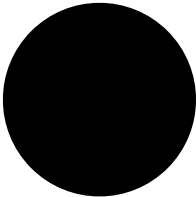


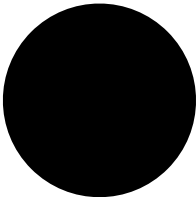
2

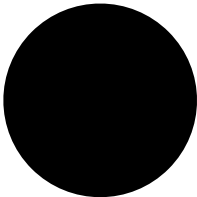


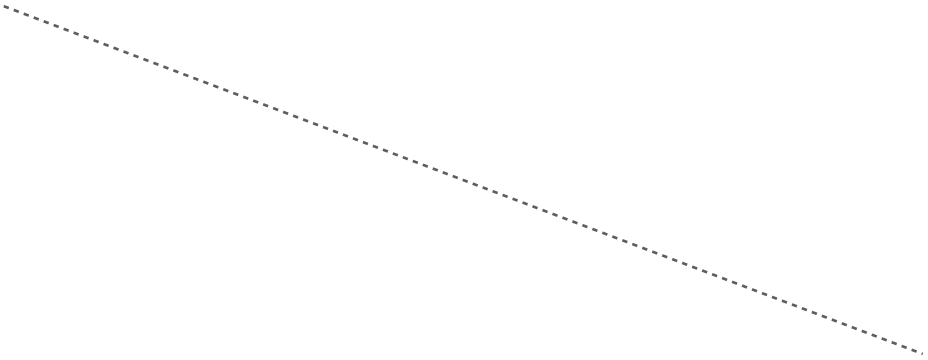




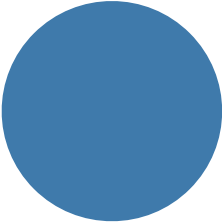


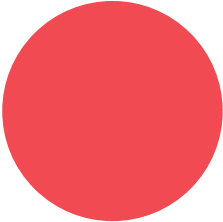


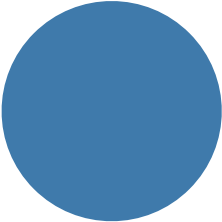


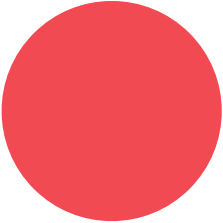


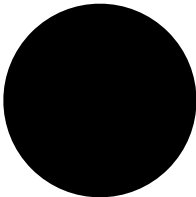












11

$x_1^{l-1} f$

1

22

$x_2^l f$

2

12

Constructing J

$$J(X) = \frac{\partial f}{\partial X}(X)$$

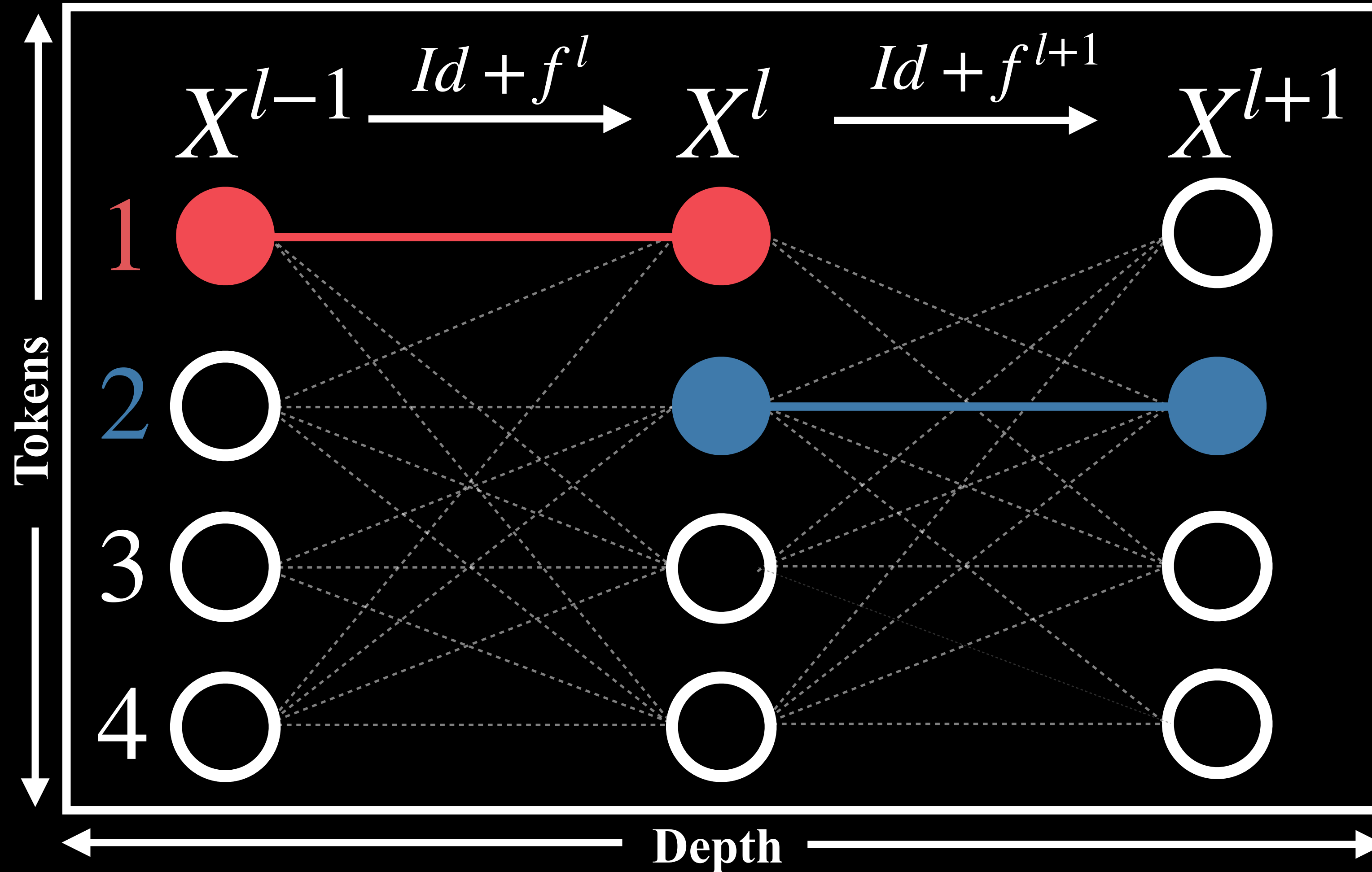
Constructing A

$$J = USV^T \quad \tilde{J} = \tilde{U}\tilde{S}\tilde{V}^T$$

$$A = \tilde{U}^T J \tilde{V}$$

Depth-wise Coupling

$$\left. \begin{aligned} J_{11}^l &= \frac{\partial}{\partial x_1^{l-1}} (f^l(X^{l-1}))_1 \\ J_{11}^{l+1} &= \frac{\partial}{\partial x_1^l} (f^{l+1}(X^l))_1 \end{aligned} \right\} A_{11}^{l,l+1}$$



Token-wise Coupling

Input = Output

$$\left. \begin{aligned} J_{11}^l &= \frac{\partial}{\partial x_1^{l-1}} (f^l(X^{l-1}))_1 \\ J_{22}^{l+1} &= \frac{\partial}{\partial x_2^l} (f^{l+1}(X^l))_2 \end{aligned} \right\} A_{11}^{l,l+1}$$

Constructing J Constructing A

Depth-wise Coupling

Token-wise Coupling

Input $\neq$ Output

$$\begin{aligned}
 J_{12}^l &= \frac{\partial}{\partial x_1^{l-1}} (f^l(X^{l-1}))_2 \\
 J_{34}^{l+1} &= \frac{\partial}{\partial x_3^l} (f^{l+1}(X^l))_4
 \end{aligned}
 \left. \vphantom{\begin{aligned} J_{12}^l \\ J_{34}^{l+1} \end{aligned}} \right\} A_{12}^{l,l+1}$$

Tokens

Depth